PRACTICAL NO. 3

Aim: Create the cube with suitable dimension and fact tables based on ROLAP, MOLAP and HOLAP model.

Tools Used:

Microsoft SQL Server 2022

SQL Server Management Studio (SSMS)

Theory:

ROLAP (Relational OLAP):

- **Data is stored in relational tables**
- **Queries are performed using SQL**
- **Suitable for large datasets**

MOLAP (Multidimensional OLAP):

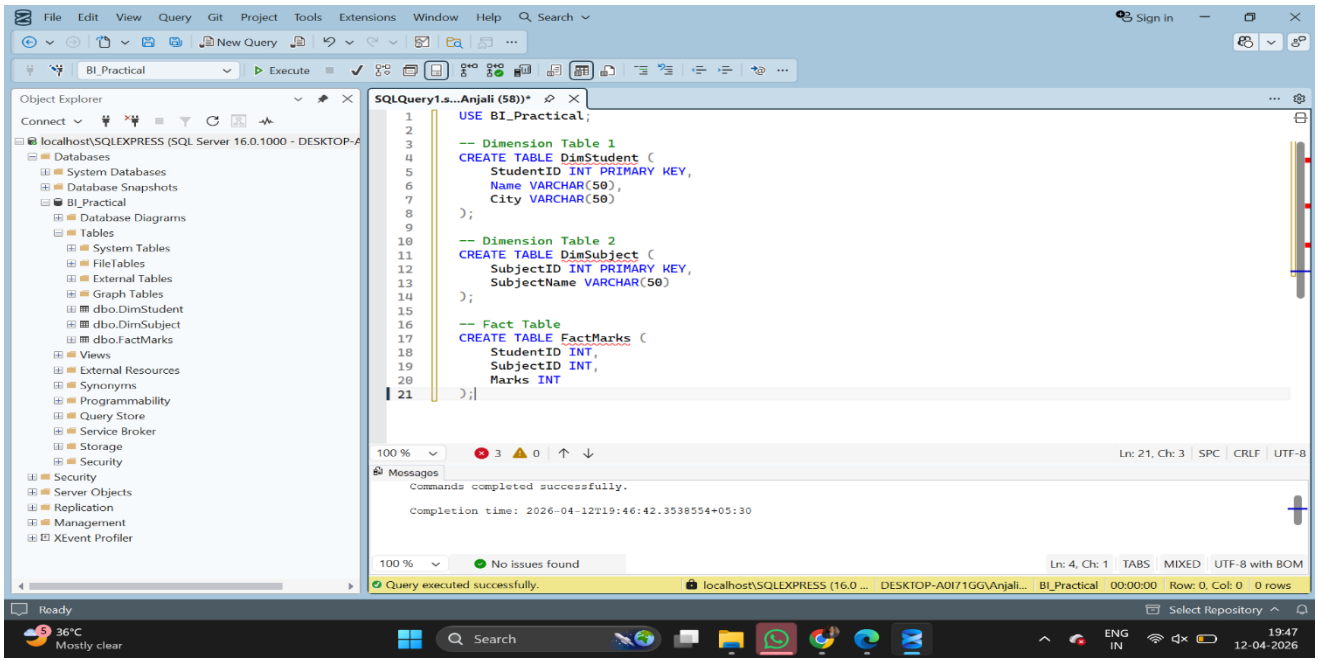
- **Data is stored in multidimensional cube**
- **Fast query performance due to pre-computation**
- **Requires OLAP tools like SQL Server Analysis Services**

HOLAP (Hybrid OLAP):

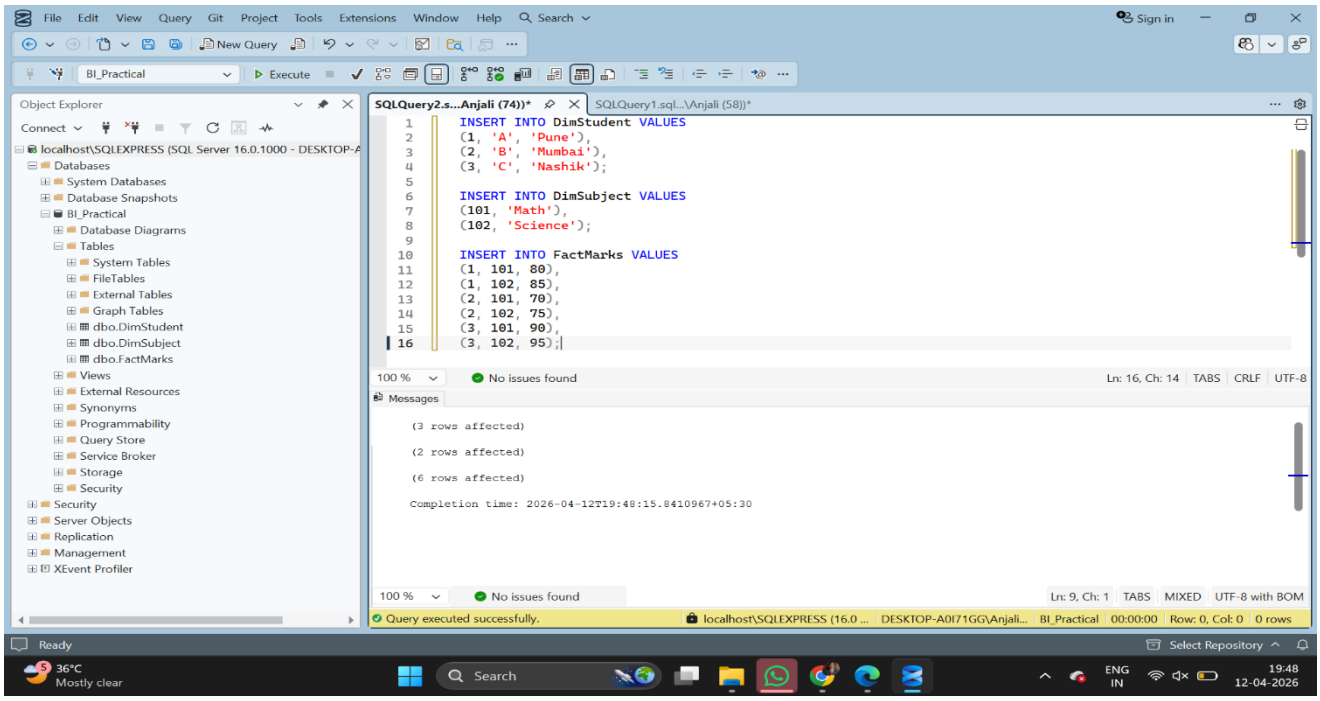
- **Combination of ROLAP and MOLAP**
- **Detailed data in relational tables**
- **Aggregated data stored in cube**

Procedure:

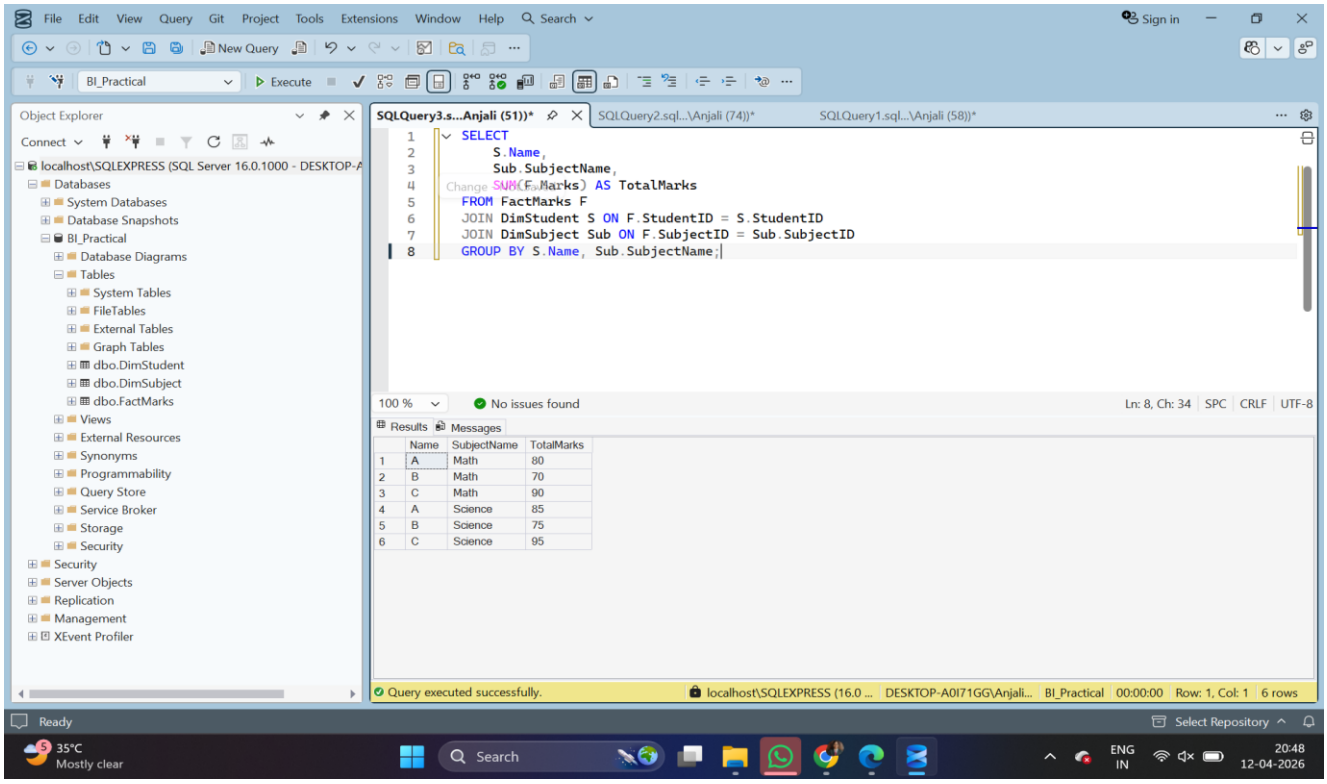
Step 1: Create Dimension and Fact Tables



Step 2: Insert Data into Tables



Step 3: Perform OLAP Operation (ROLAP Query)



Result:

- **Fact and Dimension tables were successfully created.**
- **Data was inserted successfully.**
- **OLAP operation was performed using SQL query (ROLAP).**
- **Output shows aggregated marks based on student and subject.**

Conclusion:

- **ROLAP model was implemented using SQL queries.**
- **MOLAP and HOLAP models were studied theoretically.**
- **Cube creation requires additional tools like Analysis Services.**